

Ethan Moon

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Education

University of Maryland, College-Park

August 2026

Bachelor of Science in Information Science

Experience

AI and Robotics Research Aide, Argonne National Laboratory

July 2026 – Now

- Developed and debugged a robotic arm simulation and motion planning pipeline for a UR5 manipulator using MoveIt2, Gazebo, RViz, and Python and tuning RRT* path planning algorithms to achieve reliable end-to-end trajectory execution

Data Engineer Intern, Pacific Northwest National Laboratory (PNNL)

Jun 2024 – Aug 2024

- Reduced log ingestion volume by 45% (2 TB → 1.1 TB) by eliminating redundant fields across ingestion pipelines, delivering \$500K+ in projected annual cost savings
- Designed scalable daily data pipelines processing 2 TB/day while enforcing data retention policies and operational constraints
- Built a data value scoring framework to prioritize high-utility fields and guide system-wide filtering decisions via tableau dashboards
- Analyzed ingestion patterns using Splunk (SPL) and SQL to support observability-driven optimization and inform stakeholder decisions

Computational Science & AI Intern, Fermi National Accelerator Laboratory

June 2023 – May 2024

- Automated generation and filtering of 18,000+ simulation logs using backend Python scripts, reducing manual parsing overhead
- Profiled and optimized Python backend code to diagnose performance bottlenecks in terabyte-scale data ingestion pipelines, reducing compute overhead and improving throughput
- Conducted root cause analysis on distributed ingestion workflows using logging, metrics, and controlled experiments to identify sources of latency and throughput degradation
- Collaborated with platform engineers on GlideinWMS to support distributed job scheduling and workload orchestration across compute clusters

AI and Robotics Intern, Argonne National Laboratory

June 2022 – August 2022

- Designed sensor-driven control systems leveraging inverse kinematics to enable adaptive robotic behavior in dynamic environments
- Automated robotic testing workflows using Lua scripting, improving simulation repeatability and reducing manual testing overhead

SKILLS

Languages: Python, SQL, JavaScript, Lua

Data Engineering: ETL/data pipelines, high-volume data ingestion (TB/day), ingestion optimization, data retention policies

Analytics & Observability: Splunk (SPL), Tableau, log analysis, field extraction and reduction

Robotics: NVIDIA Isaac Sim, Gazebo, ROS 2, CAD

Activities and Presentations

DOE Cybersecurity and Technology Innovation Conference 2024

Aug 2024

- Presented Splunk data management solutions to cybersecurity professionals and industry leaders.